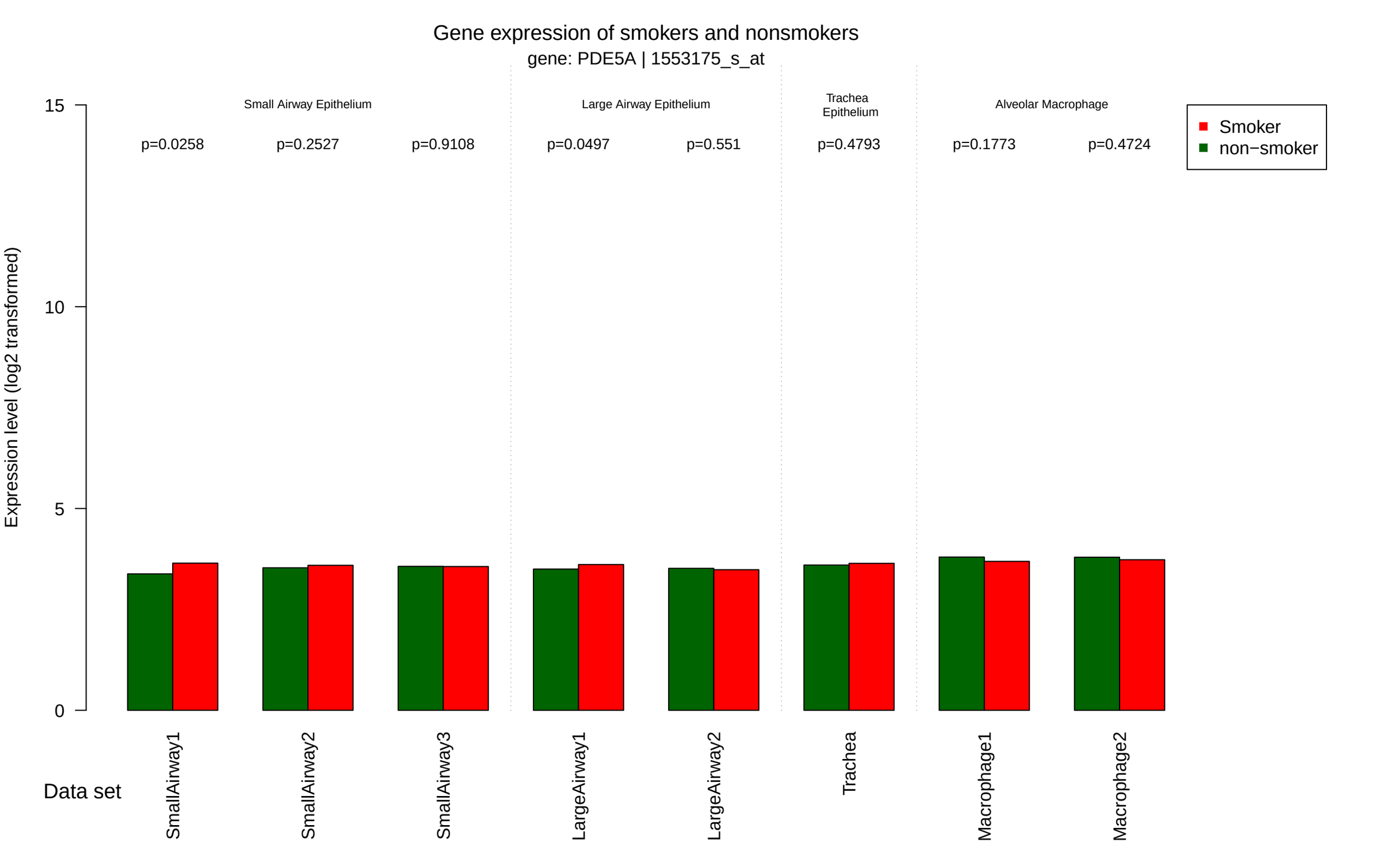
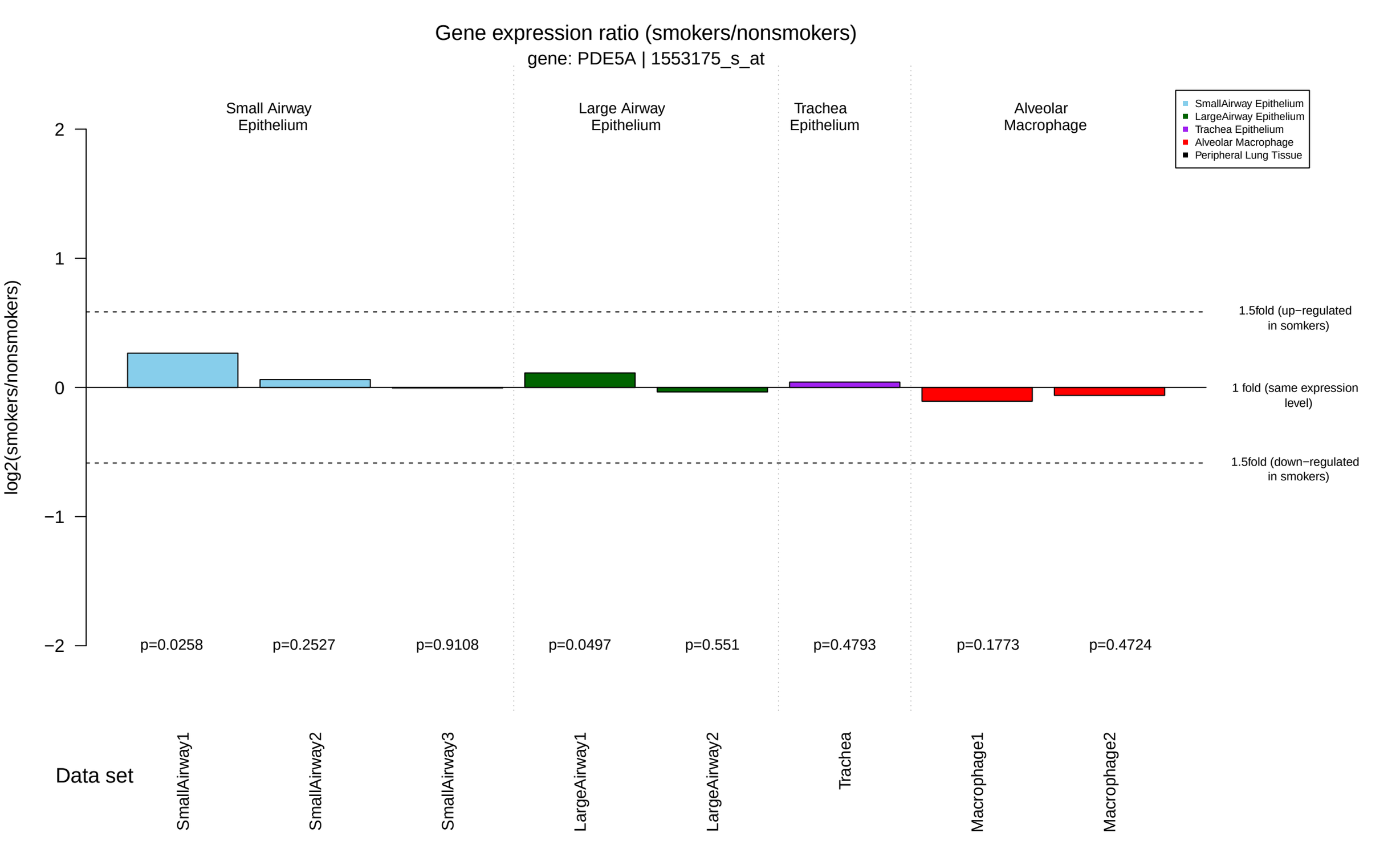
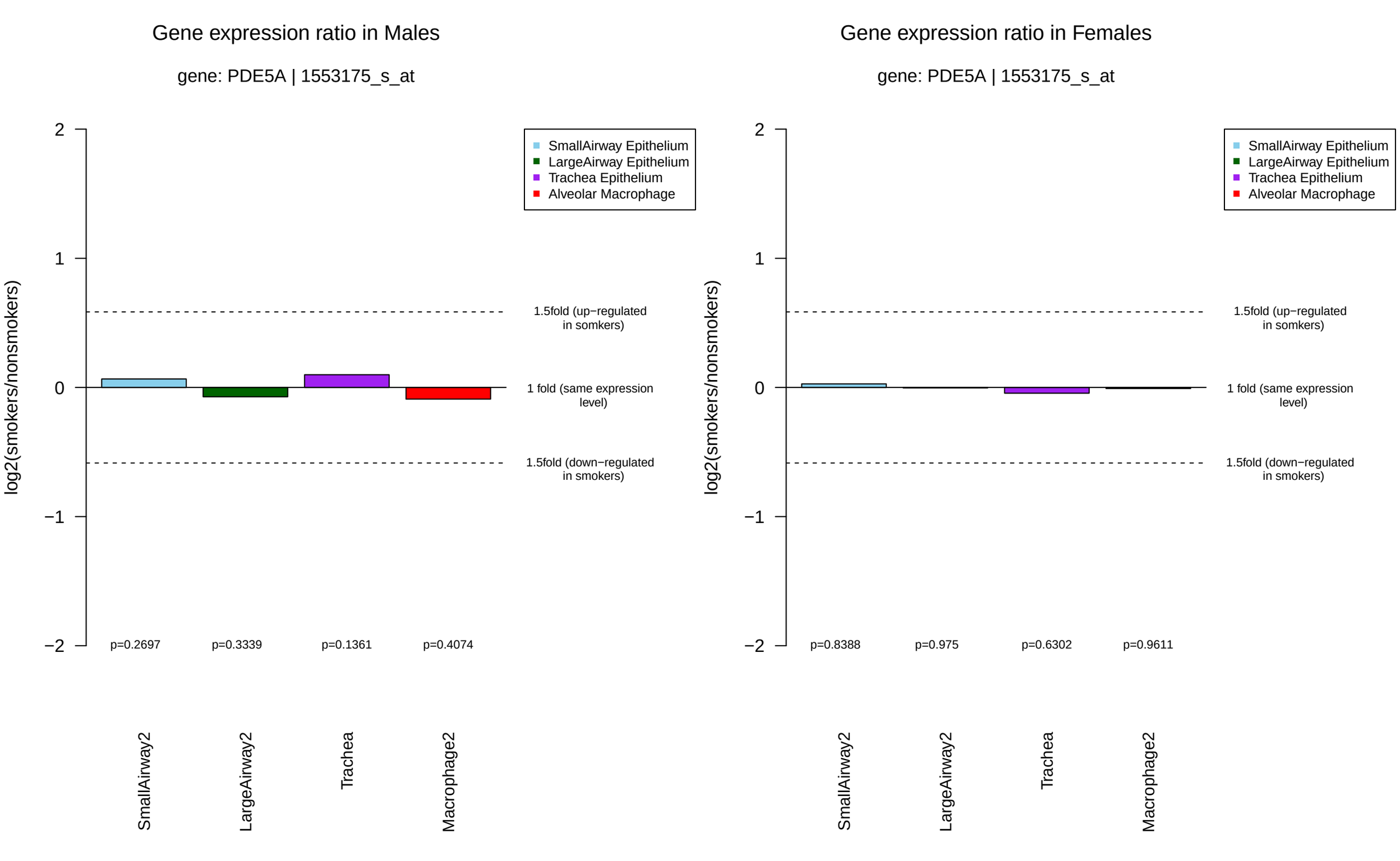
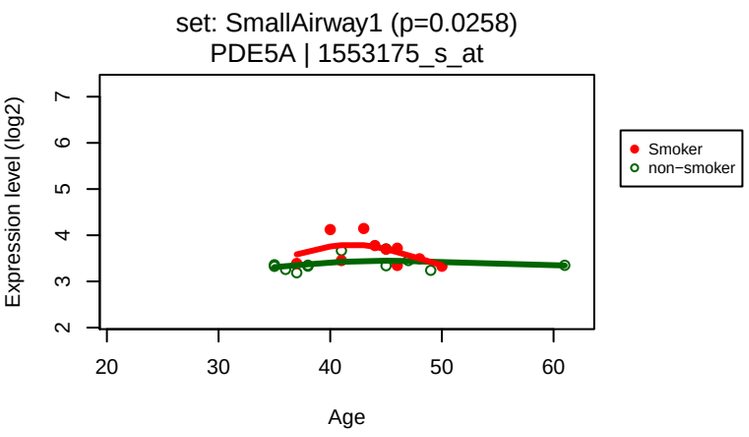
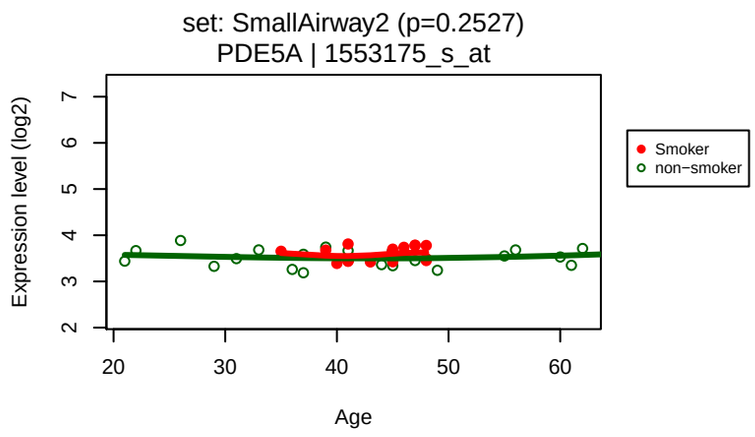
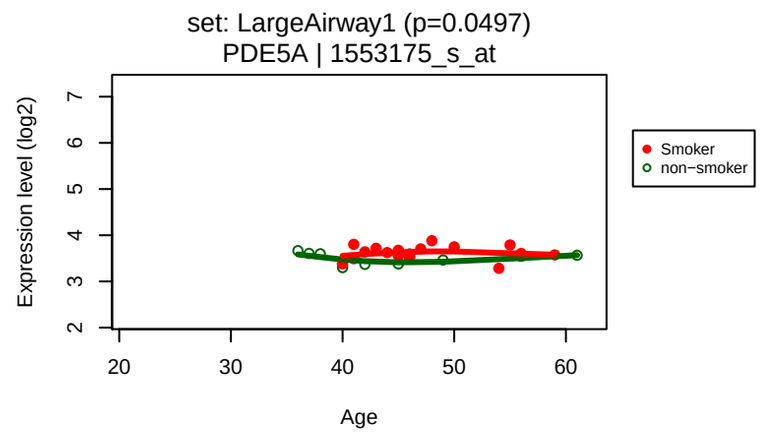

Gene expression of smokers and nonsmokers
gene: PDE5A | 1553175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0258
p=0.2527
p=0.9108
p=0.0497
p=0.551
p=0.4793
p=0.1773
p=0.4724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

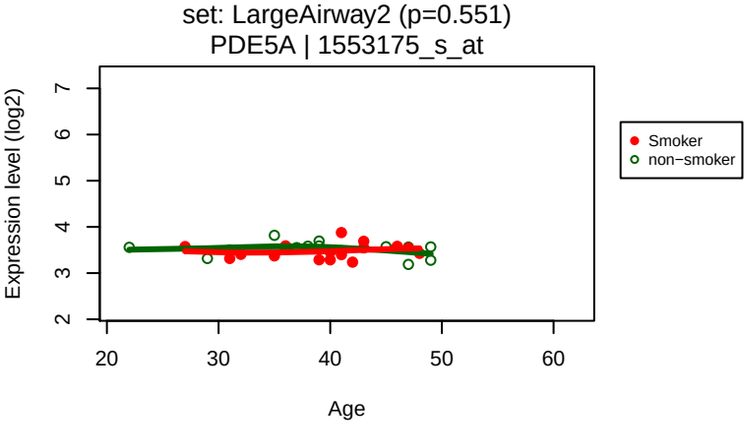

Gene expression ratio (smokers/nonsmokers)
gene: PDE5A | 1553175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0258
p=0.2527
p=0.9108
p=0.0497
p=0.551
p=0.4793
p=0.1773
p=0.4724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

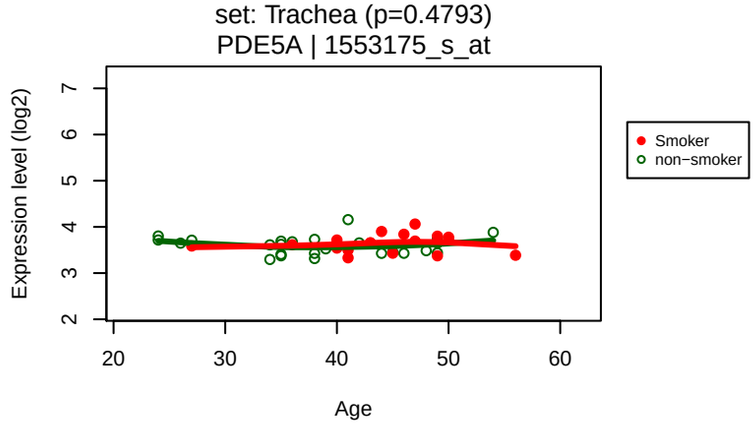

Gene expression ratio in Males
gene: PDE5A | 1553175_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2697
p=0.3339
p=0.1361
p=0.4074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE5A | 1553175_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8388
p=0.975
p=0.6302
p=0.9611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

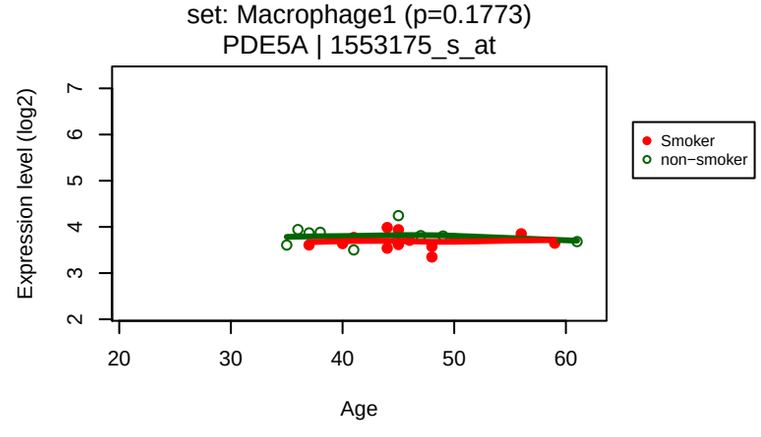

set: SmallAirway1 (p=0.0258)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker

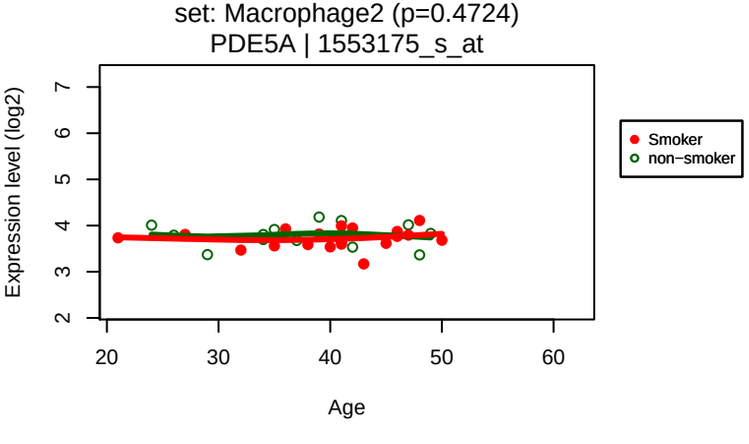

set: SmallAirway2 (p=0.2527)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker

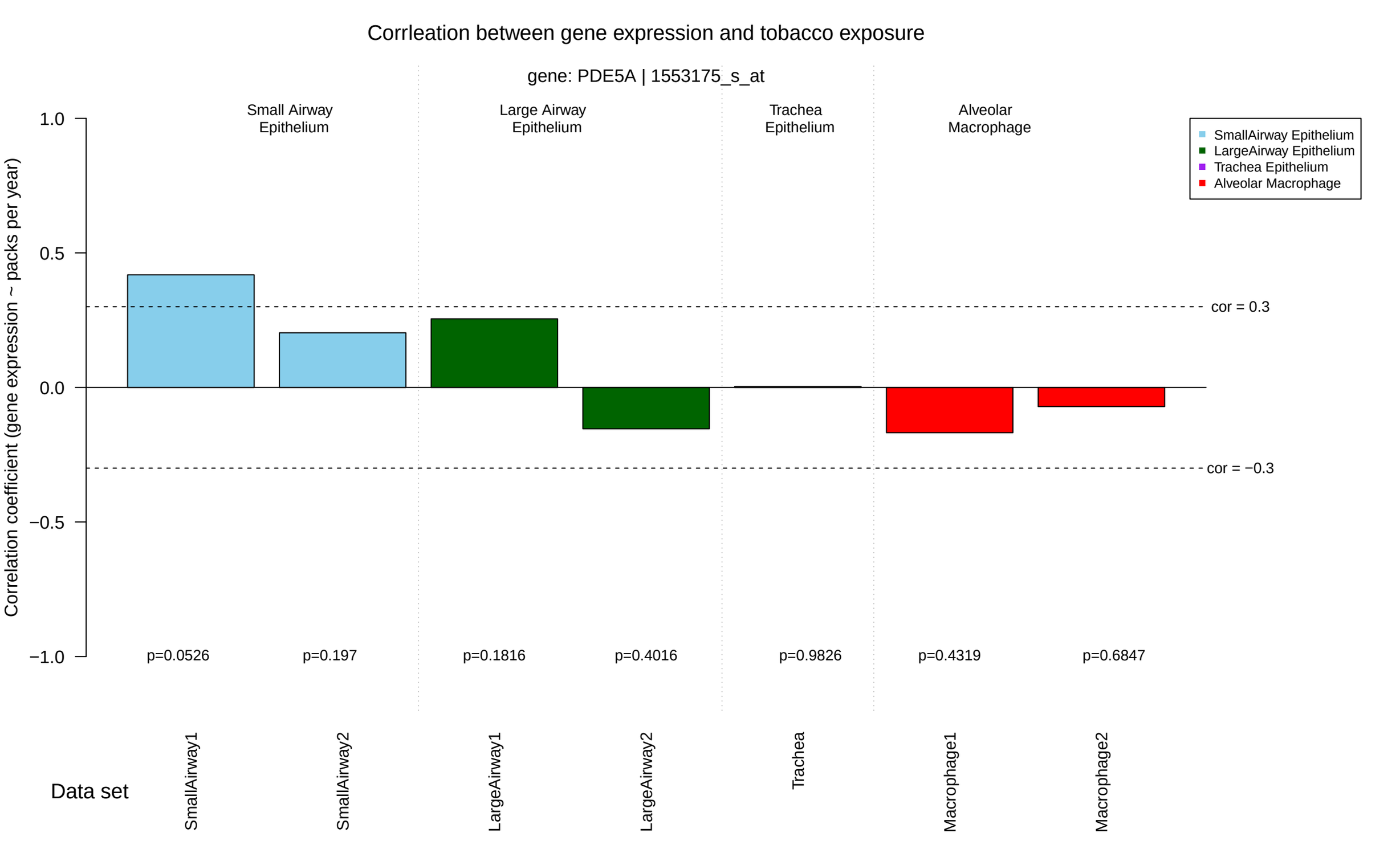

set: LargeAirway1 (p=0.0497)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.551)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4793)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1773)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4724)
PDE5A | 1553175_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE5A | 1553175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0526
p=0.197
p=0.1816
p=0.4016
p=0.9826
p=0.4319
p=0.6847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2